

The big picture

Guiding question(s)

What determines the covalent nature and properties of a substance?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Everything around you has been carefully selected for its use on the basis of its properties. Look at the items in **Figure 1**. If there is a fire extinguisher nearby you, then it is filled with carbon dioxide, chosen because it can be used to quench fires. The pencil you write with is probably made of graphite, because of its ability to leave an impression on the page even when you apply very little force. The computer keyboard you are typing on is probably made of a type of plastic, selected for its durability and lightness.

Consider what would happen if each of the uses of these materials was reversed. Would graphite be suitable on a computer keyboard? Could carbon dioxide be used in pencils or plastic in fire extinguishers?

Credit: boschettophotography,

Getty Images

Credit: wavebreakmedia, Getty Images

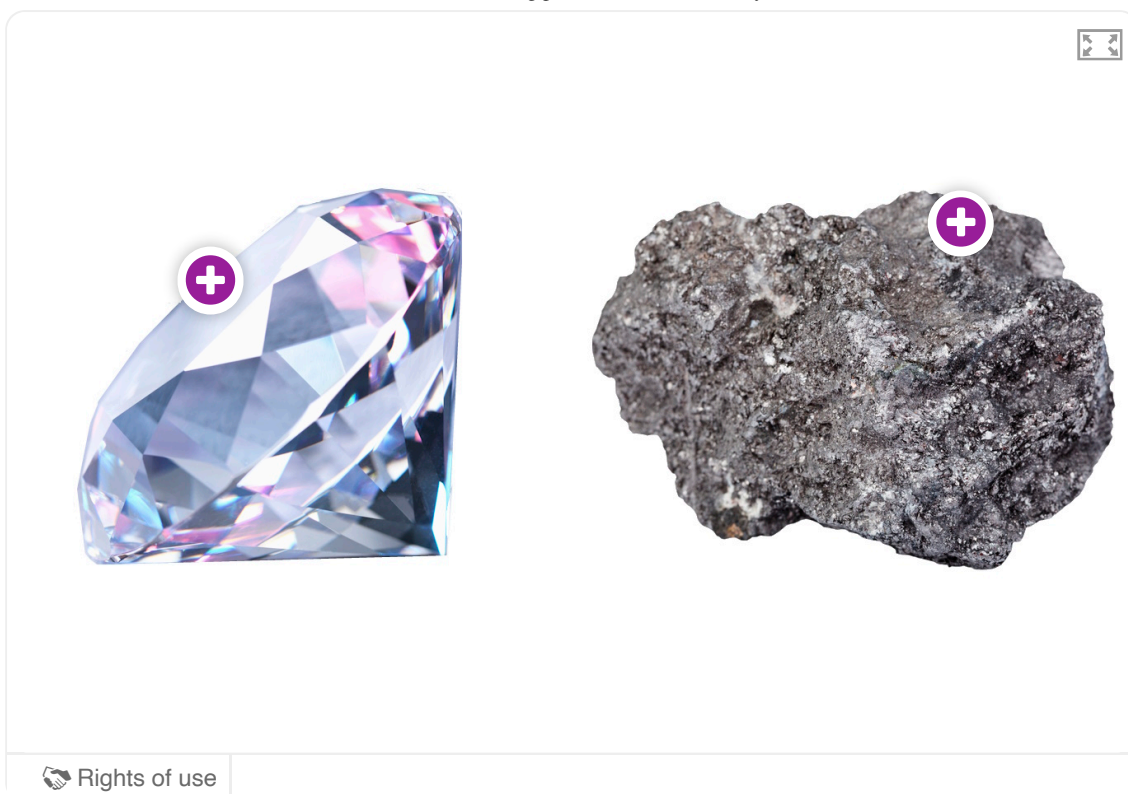
Credit: Image Source, Getty Images

Figure 1. The atoms, and how they are bonded to each other, determined the properties of each of these materials.

The uses of these materials are not interchangeable and although they have quite different properties they all contain the same element, carbon. Carbon forms a type of bond known as a covalent bond, a different type of bond from the ionic bonds you learned about in section S2.1.1 (<https://app.kognity.com/:sectionlink:123789>). Carbon can form these covalent bonds with itself, for example in graphite, and with other elements, such as in carbon dioxide. Changing the element carbon has bonded with affects the properties the substance has but also alters the structure of the material itself. The type of bond being formed is also important as this will determine its properties and hence the properties of the overall substance.

Consider another carbon-containing compound, diamond. Although both graphite and diamond are both made of only carbon atoms, they have wildly different properties. Graphite is soft enough to slide across a page, leaving a line. Diamond is one of the hardest substances known to humankind, used in cutting and drilling machines. Diamond is rarer on Earth than graphite, which also makes it significantly more valuable.

Look at the structure of diamond and graphite in **Interactive 1**.



Interactive 1. The Structure of Graphite and Diamond Determines its Properties.

[More information for interactive 1](#)

This subtopic will explore the structure of materials such as diamond, graphite, plastic and molecules such as carbon dioxide. It will also explore how structure determines the reactivity and properties of materials.

Prior learning

Before you study this subtopic make sure you understand the following:

- How to draw and write the electron configuration of an element (see [section S1.3.5 \(https://app.kognity.com/:sectionlink:124608\)](https://app.kognity.com/:sectionlink:124608)).
- Describe the properties of ionic compounds (see [section S2.1.3a \(https://app.kognity.com/:sectionlink:124596\)](https://app.kognity.com/:sectionlink:124596)).